



East West University

Lab Report

Semester: Summer-2025

Course Title: Electronic Circuits **Course Code:** CSE251

Section: 04

Expt No: 06

Expt Name: Signal Integration and Differentiation using 741 Op Amp

Group No: 06

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Date of Performance: August 11, 2025

Date of Submission: August 18, 2025

Title: Signal Integration and Differentiation using 741 Op Amp

Objectives

1. To study the response of 741 Op Amp Integrator to sinusoid and square waveforms.
2. To study the response of Op-Amp differentiators to sinusoid and triangular waveforms.

Theory (Summary)

An operational Amplifier (Op-Amp) is a differential amplifier that performs mathematical operations such as addition, subtraction, integration, differentiation, etc. In Expt. No. 3, we use the Op-Amp with the only resistors to study adder and amplifier circuits. In this experiment, we will study integrator and differentiator circuits by using a capacitor either in the feedback or input path.

Equipment and Component

1. Digital trainer board
2. Signal generator
3. Oscilloscope
4. Digital multimeter
5. 741 Op Amp (1 pc)
6. Resistor ($1\text{K}\Omega$ 1 pc)
7. Capacitor ($0.1\mu\text{F}$, 1 pc)
8. Breadboard
9. Connecting wires

Circuit Diagram

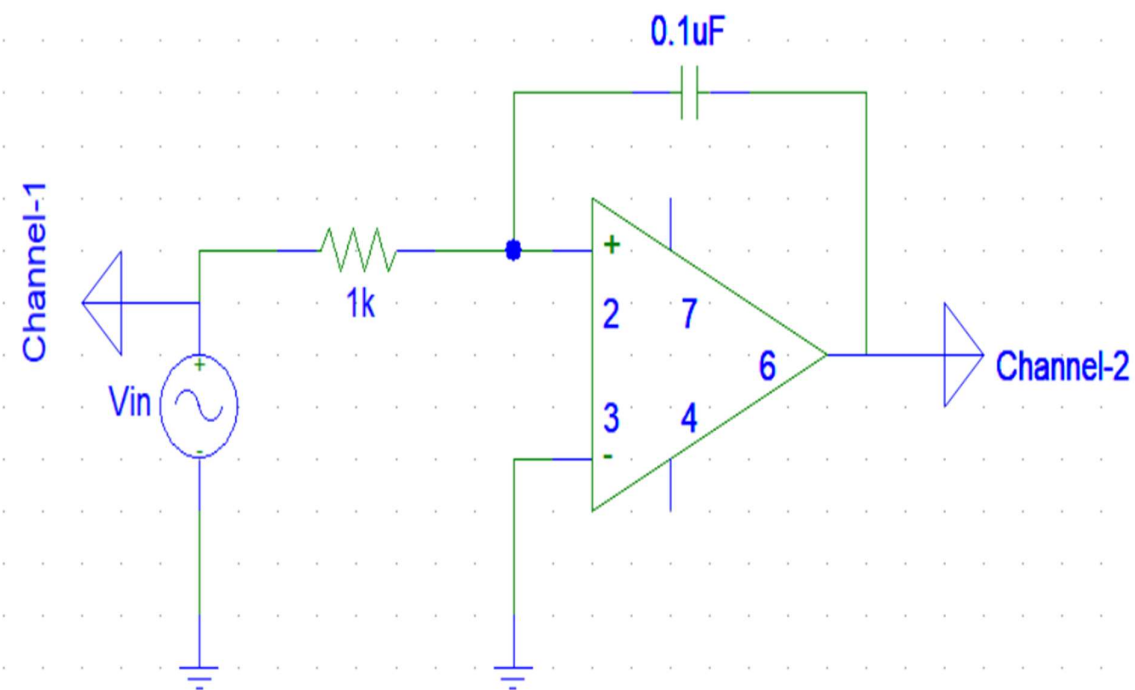


Figure-1: An Op Amp integrator circuits.

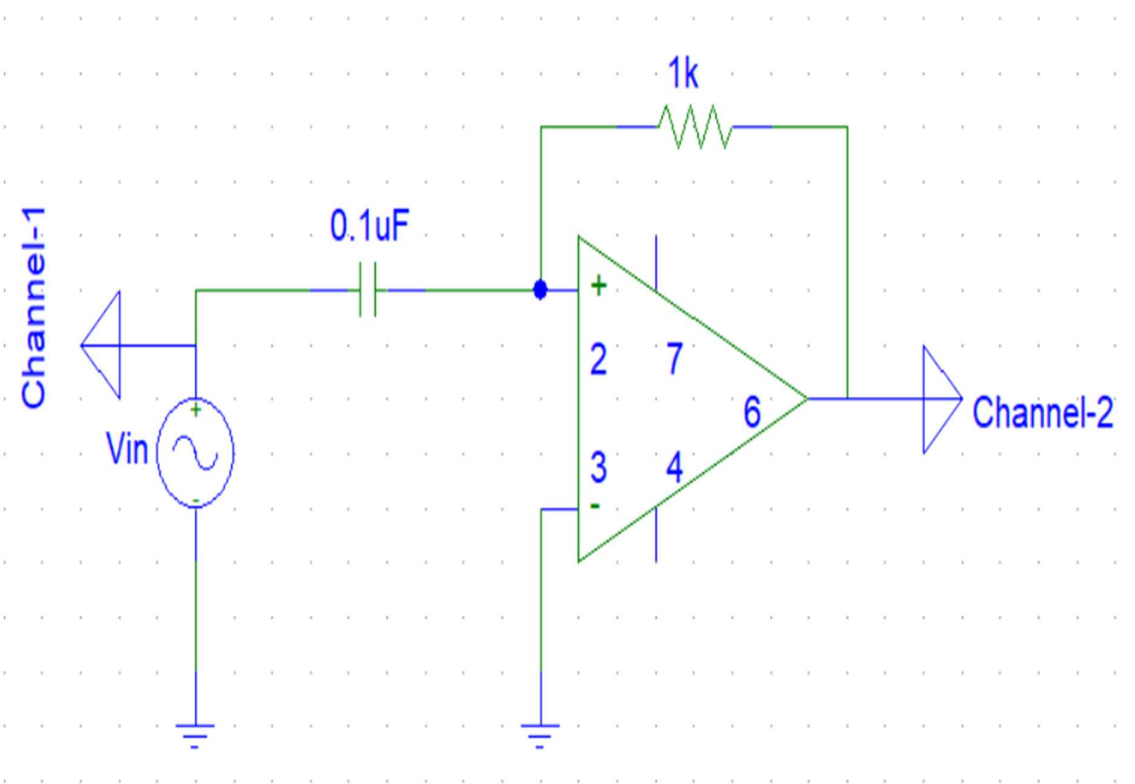


Figure-2: An Op Amp differential circuits

Pre-Lab

1. INTEGRATOR: Consider the integrator shown in Figure_1 with $V_{in} = V_P \sin(\omega t)$. If the circuit is initially relaxed, find the expression of output and determine the frequency, f , at which the output amplitude is equal to $2V_P$. Also, determine the phase relation between input and output.

Answer

Integrator Circuit Given

Input: $V_{in} = V_P \sin(\omega t)$

Resistor: $R = 1K\Omega = 1000\Omega$

Capacitor: $C = 0.1\mu F = 1 \times 10^{-7}F$

Output Expression:

For an ideal op-amp integrator, the output is:

$$V_{out}(t) = -\frac{1}{R_c} \int V_{in}(t) dt$$

Substitute, $V_{in}(t) = V_P \sin(\omega t)$

$$V_{out}(t) = -\frac{V_P}{RC\omega} \cos(\omega t)$$

So, the output is a cosine wave with a phase shift of -90° (lagging input by 90°), and

amplitude $V_{out} = \frac{V_P}{RC\omega}$

Now when output amplitude = $2V_P$

$$\therefore \frac{V_P}{RC\omega} = 2V_P$$

$$\text{Or, } 2\pi f = \frac{1}{2RC}$$

$$\text{Or, } f = \frac{1}{4\pi RC} = \frac{1}{4\pi \times 1 \times 10^{-7} \times 1000} = 795.7747 \text{ Hz}$$

Phase Relation; Output lags the input by 90°

2. DIFFERENTIATOR: Consider the differentiator shown in Figure 2 with $V_{in} = V_p \sin(\omega t)$. If the circuit is initially relaxed, find the expression of output and determine the frequency, f , at which the output amplitude is equal to $V_p/2$. Also, determine the phase relation between input and output.

Answer

Differentiator Circuit Given

Input: $V_{in} = V_p \sin(\omega t)$

Resistor: $R = 1K\Omega = 1000\Omega$

Capacitor: $C = 0.1\mu F = 1 \times 10^{-7}F$

Output Expression:

For an ideal op-amp integrator, the output is:

$$V_{out}(t) = -RC \frac{d}{dt} V_{in}(t)$$

Substitute, $V_{in}(t) = V_p \sin(\omega t)$

$$V_{out}(t) = -\omega RC V_p \cos(\omega t)$$

So, the output is a cosine wave with a phase-shifted -90° (lagging input by 90°), and amplitude, $V_{out} = -\omega RC V_p$

Now when output amplitude $= \frac{V_p}{2}$

$$\therefore \omega RC V_p = \frac{V_p}{2}$$

$$\text{Or, } 2\pi f = \frac{1}{2RC}$$

$$\text{Or, } f = \frac{1}{4\pi RC} = \frac{1}{4\pi \times 1 \times 10^{-7} \times 1000} = 795.7747 \text{ Hz}$$

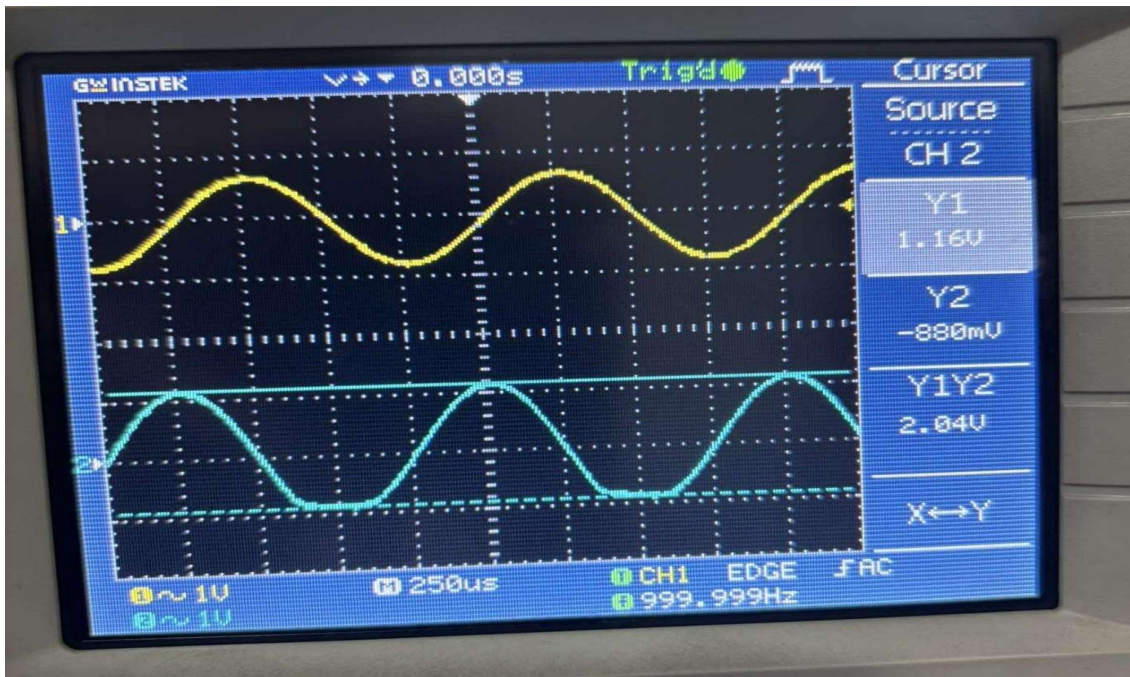
Phase Relation: Output lags the input by 90°

Post-Lab

INTEGRATOR:

1. In the output expression from your prelab, put the measured value of R and the given values of C and V_P ; calculate the amplitude of the output signal and compared it with the measured data.

Answer



Here,

$$R = 1K\Omega = 1000\Omega$$

$$C = 0.1\mu F = 1 \times 10^{-7}F$$

$$V_P = 2V$$

$$f = 1000Hz$$

We know,

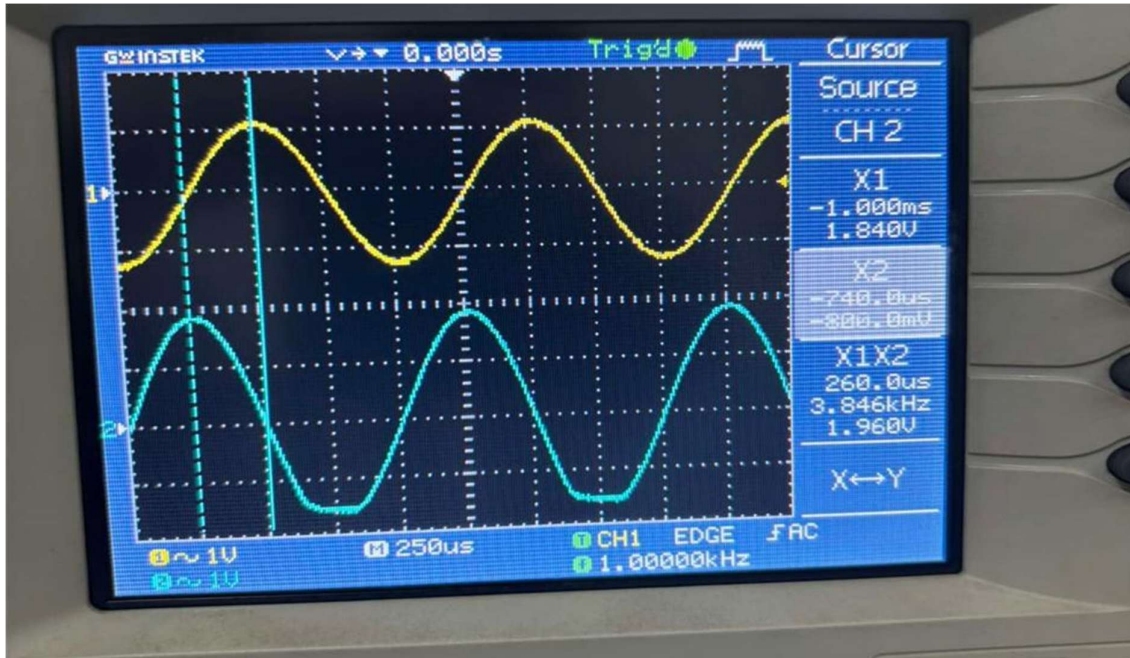
$$\text{Amplitude, } V_o = \frac{V_P}{\omega RC} = \frac{V_P}{2\pi f RC} = \frac{1}{2\pi \times 1000 \times 1000 \times 1 \times 10^{-7}} = 3.18V$$

Comparison between the measured & calculate the amplitude:

	Calculate Value	Measured Value	Discrepancy
Amplitude	3.18	2.04	35.849%

2. Compare the phase relation between your prelab result and measured data.

Answer



Here,

$$\Delta t = 260\mu s$$

$$f = 1000\text{Hz}$$

$$\begin{aligned}\therefore \Delta\theta &= 360^\circ \times \Delta t \times f \\ &= 360^\circ \times 260\mu s \times 1000 = 93.6^\circ\end{aligned}$$

Comparison between the measured & prelab values of the phase difference:

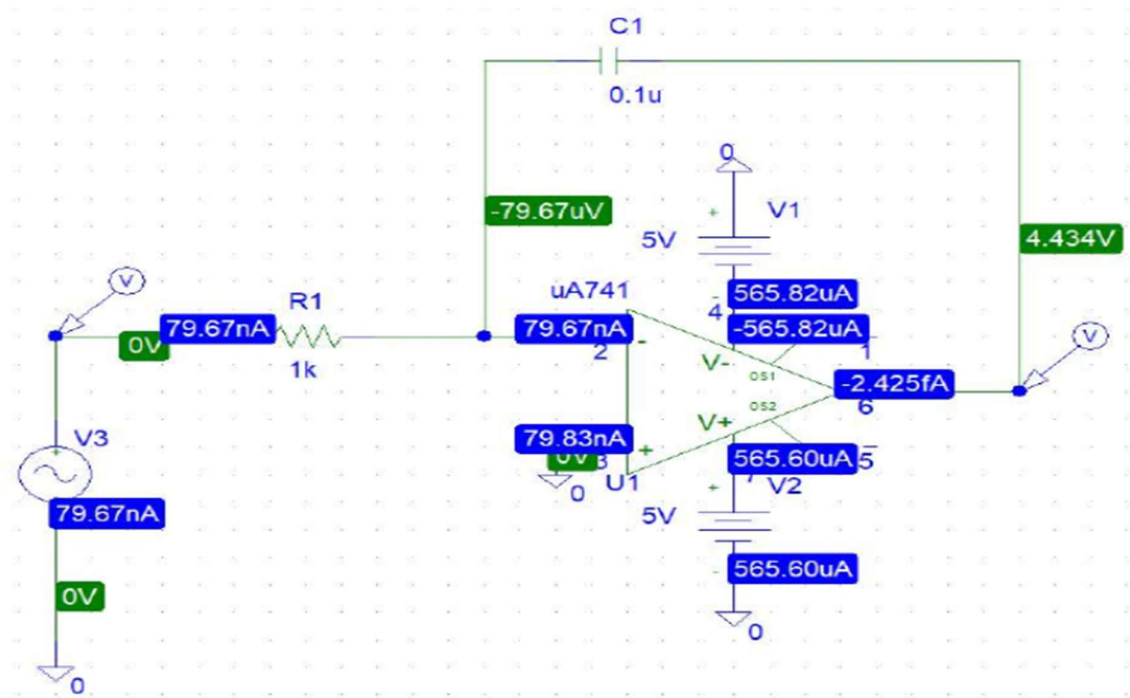
	Prelab Value	Measured Value	Discrepancy
Phase Relation	90°	93.6°	4%

3. Simulate the integrator of Figure-1 for 4 cycles using PSpice with the measured value of R and the square wave input that you have used in the lab and have the simulation result printed. Compare the simulation result with your measurement.

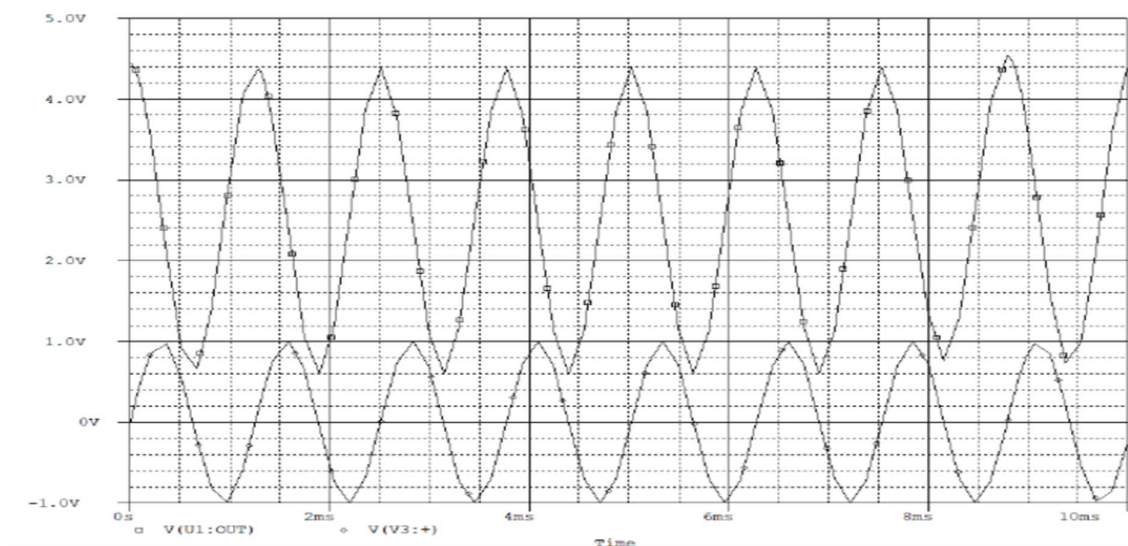
Answer

Operation of Op Amp Integrator:

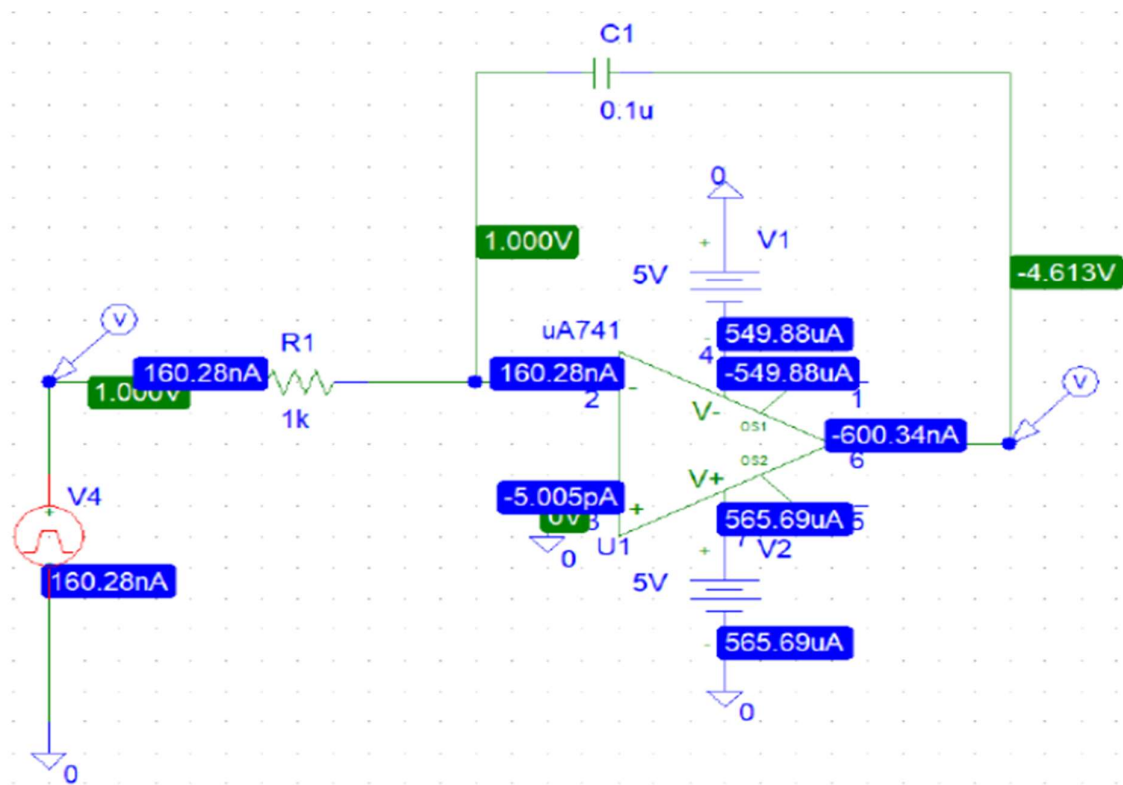
Integration of Sinusoidal Voltage:



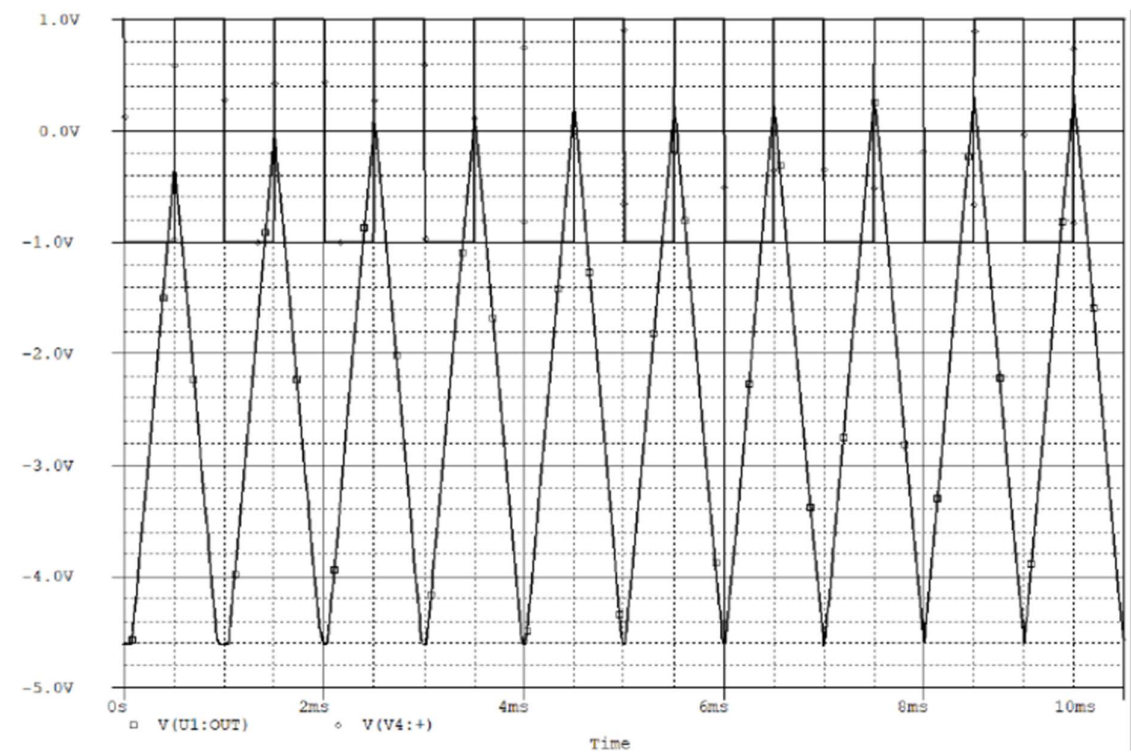
Simulation Result:



Integration of Square wave Voltage:



Simulation Result:



Discussion

This experiment successfully demonstrated the use of the 741 Op Amp to perform basic analog signal processing functions - integration and differentiation. Also in different waveforms like square, sine, cosine and triangular according to calculus principles.

In the integrator circuit when a square wave was applied as input, the output was a triangular wave indicating the time integration of the signal. Similarly, a sine wave signal input produced a phase shifted cosine.

The measured data showed a notable discrepancy with the pre-lab calculation for the integrator circuit in amplitude and a much smaller in phase relation, may be due to several non-ideal characteristics of the 741 Op Amp and the components, the limited bandwidth at the operating frequency, Oscilloscope measurements and circuit design. The measured frequency was also close to the calculated frequency. However, the PSpice simulation were consistent with the theory.

In conclusion, While there were discrepancies, the experiment successfully provided practical insight into the behavior of Op Amp based integrator and differentiator circuits and their applications in waveform shaping, analog computation and signal processing.